

Krutika Jagdale

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PROFESSIONAL SUMMARY

Security engineer with 4+ years in detection engineering, threat analysis, and incident response across enterprise SOC environments. Grounded in fundamentals - **SIEM, Python automation, supply-chain security, and cloud security** - with a genuine curiosity for how emerging technology reshapes the threat landscape. Quick to learn new tools and systems; focused on turning emerging threats into practical security solutions.

TECHNICAL SKILLS

Detection Engineering: MITRE ATT&CK mapping, Sigma/YARA, custom correlation rules, detection tuning, threat hunting
SIEM, EDR and Incident Response: Splunk (SPL), IBM QRadar (AQL), Microsoft Sentinel (KQL), CrowdStrike EDR/XDR, Snort and Suricata IDS/IPS, incident triage and containment, digital forensics, Wireshark
Application Security: Semgrep, Checkmarx, Snyk, Burp Suite, Nmap, Nessus, Tenable, OWASP Top 10, STRIDE threat modeling
Supply-Chain and AI Security: OSV.dev, MCP security scanning, prompt injection defense, OWASP LLM Top 10
Cloud and Identity Security: AWS (IAM, EC2, S3, VPC, CloudTrail, GuardDuty), Azure Defender, GCP, Docker, Kubernetes, Terraform, Active Directory, Entra ID, SAML, OAuth 2.0, MFA, RBAC, Zero Trust
Network, Programming and Frameworks: TCP/IP, DNS, TLS, IPsec VPN, Palo Alto and Cisco ASA firewalls, Python (pytest, Ruff, Mypy), Bash, PowerShell, Ansible, SQL, Git, REST APIs, CI/CD, Linux/RHEL, NIST CSF, ISO 27001, SOC 2

PROFESSIONAL EXPERIENCE

IBM <i>Security Engineer - SIEM and Detection Engineering</i>	Pune, India Aug 2021 - Aug 2024
- Monitored 20+ enterprise SOC environments at 10K-50K EPS; investigated and resolved 1,000+ security incidents. - Engineered 40+ MITRE ATT&CK-mapped detection rules in Splunk and QRadar; reduced MTTD by 35%. - Retuned noisy detections found during case review; cut false positives 45% and improved analyst signal quality. - Developed SIEM detection use cases from TTPs and IOCs across endpoint, network, and cloud log sources. - Remediated 80%+ critical vulnerabilities across QVM, QRM, and QNI; tracked fixes with owners to closure. - Maintained 95% SLA and 99.99% availability for 4-5 accounts; briefed technical and executive stakeholders.	
College of Liberal Arts and Sciences, University of Colorado Denver <i>IT Associate - Systems, Identity and Security Automation</i>	Denver, CO Mar 2025 - Dec 2025
- Administered identity management and authentication (Active Directory, Entra ID, MFA, RBAC) for 500+ users. - Troubleshoot system, network, and access issues using command-line tools; resolved 150+ monthly issues at 95% SLA. - Built Python and PowerShell automation and API integrations; reduced manual operational overhead by 70%.	
Digital4Good and ICANHELP <i>Security Analyst Intern</i>	Remote, CA Aug 2025 - May 2026
- Triaged 7 phishing datasets and flagged 98 adversarial emails through structured IOC classification. - Identified recurring attacker infrastructure patterns across 3 datasets; briefed team on threat actor TTPs. - Authored threat communications translating technical findings into plain-language guidance for non-security staff.	

PROJECTS

SupplyX: AI-Native Software Supply-Chain Security Scanner - <i>Python, Pydantic, OSV.dev, MCP, SARIF, CycloneDX, pytest</i>
- Parsed <code>pyproject.toml</code>, <code>uv.lock</code>, and <code>requirements.txt</code> into deterministic graphs classifying direct vs. transitive packages. - Correlated packages against OSV.dev into a canonical vulnerability-to-evidence graph with 0-10 contextual risk scoring. - Built a static MCP scanner detecting five capability classes (filesystem, shell, network, credential, endpoint). - Emitted SARIF 2.1.0 and CycloneDX SBOM with deterministic IDs and secret sanitization for CI/CD gating. - Added a provider-neutral LLM layer with citation grounding; tested with pytest, Ruff, Mypy.
ThreatSight: Detection Engineering and Data Pipeline Toolkit - <i>Python, MITRE ATT&CK, Sigma, REST APIs</i>
- Built a Python agent ingesting threat intel via REST APIs, auto-generating MITRE ATT&CK-mapped Sigma rules. - Normalized cloud and endpoint log schemas into one hunting pipeline; cut manual rule authoring 60% via templated Sigma logic.

CERTIFICATIONS

CompTIA Security+ - Dec 2025	AWS Certified Solutions Architect Associate - Feb 2024
CompTIA Network+ - July 2025	IBM QRadar Deployment 7.5 Certification - Nov 2024
CCNA - Cisco Certified Network Associate - Sept 2025	Red Hat Certified System Administrator - Dec 2019
AI Agent Security Masterclass - Black Hat USA 2026	AI AppSec Robots: Real-World Agents - Black Hat USA 2026

EDUCATION

M.S. in Information Systems (Cybersecurity) , University of Colorado Denver, CO	Aug 2024 - May 2026
B.Tech. in Computer Science , Deogiri Institute of Engineering and Management Studies, India	Aug 2017 - May 2021

AWARDS AND COMPETITIONS

No.1 - ISACA Denver Case Study Competition (2026): Developed a security framework addressing deepfake threats and AI-driven risks; designed mitigation and detection strategies showcasing end-to-end threat modeling.
MWCC Cybersecurity CTF: Performed web app pen testing (SQLi, XSS, auth bypass); conducted digital forensics on file systems and memory dumps; executed OSINT and threat analysis demonstrating vulnerability research skills.
RMCCDC Cyber Defense Competition: Participated in live-fire Red Team/Blue Team engagements; monitored logs, hunted threats, and mitigated real-time cyberattacks.